

Apparel and Fashion Warehouse Design Walkthrough

Design basis, calculations, zone justification and presentation guide

This guide explains the 60 m x 50 m apparel and fashion warehouse concept in a way that can be defended in a presentation or interview. It separates three things: calculated geometry from the FreeCAD layout, planning assumptions used to size future operations, and checks that must be completed before real construction or operation.

Main conclusion

The design uses a 3,000 m² enclosed footprint with 2,370.07 m² allocated to functional zones and 629.93 m² assigned to walls, circulation and clearances. The layout adopts U-flow: two inbound docks and three outbound docks share the front façade, while receiving, storage, picking, processing, packing and dispatch follow a logical sequence.

How to use this guide

- Read Sections 1 to 4 to understand the logic behind the layout.
- Use Section 5 when explaining individual zones and their dimensions.

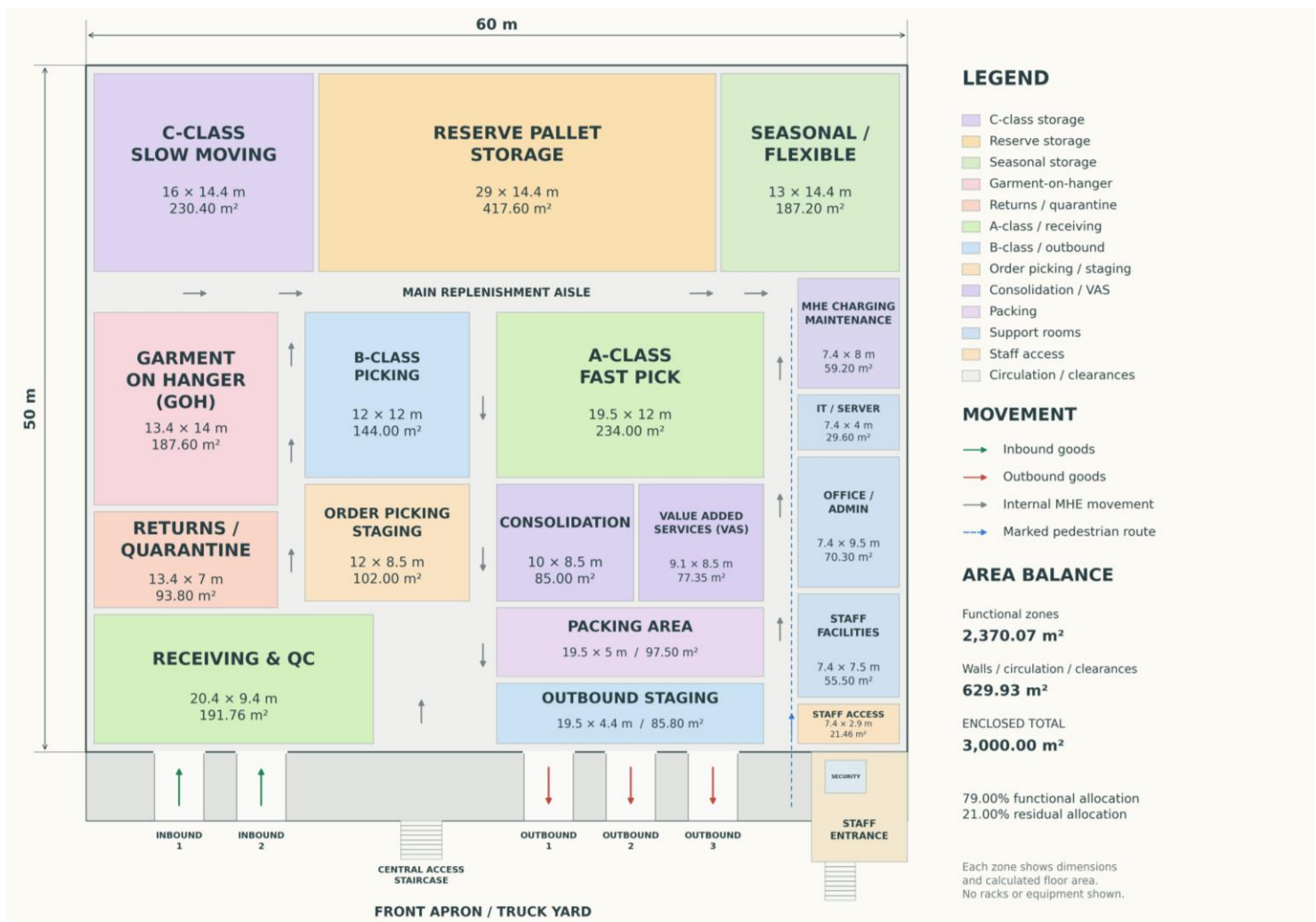


Figure 1. Dimensioned zoning plan derived from the FreeCAD concept model.

1 Design basis and evidence standard

Step	What it means in this project	Evidence available now
Requirement	Store folded garments, hanging garments, seasonal stock and returns while fulfilling orders.	Case study scope and layout requirements.
Planning input	Expected SKU mix, inbound/outbound peak volumes, pallet size, forklift type, return rate and order profile.	To be confirmed by operational data or agreed assumptions.
Calculation	Convert activity volumes and equipment needs into pallet positions, workstations, staging positions and aisle space.	Formula templates in Section 4.
Physical space	Allocate and locate each zone within the 60 m x 50 m footprint.	Exact zone dimensions in Section 3.
Validation	Check turning, rack capacity, staffing, fire safety, docks and pedestrian/MHE conflicts.	Required before implementation.

The current model is therefore a strong preliminary design. It is not a final construction drawing or safety approval.

What is verified and what is assumed

Status	Use in presentation
Verified geometry	The 60 m x 50 m footprint, zone dimensions, five dock positions, two staircases, zone total and area balance.
Planning assumption	Expected order volumes, SKU count, pallet inventory, return rate, number of packers and MHE fleet size.
Final validation item	Actual forklift turning radius, loaded aisle requirement, rack loading, fire egress, local building rules, swept truck paths and staffing.

2 Warehouse concept and flow choice

The project is an apparel and fashion warehouse. Apparel requires different storage methods because some goods arrive in cartons, some must hang to avoid creasing, order profiles contain many styles, colours and sizes, and seasonal collections can change inventory requirements quickly.

Why U-flow was chosen

Receiving and shipping use the same front façade. The two inbound docks sit on the front-left and the three outbound docks sit on the front-right. This supports one shared truck apron and makes dock activity easier to supervise. The layout does not claim that every item follows a perfectly straight route. Instead, it positions related activities in a sequence that reduces unnecessary transfers.

Operational requirement	Layout response
Receive and inspect inbound garments	Receiving and QC occupy the front-left, adjacent to inbound docks.
Protect inventory quality	Returns and quarantine sit behind receiving so rejected stock does not enter normal inventory without inspection.
Hold bulk stock and replenish pick faces	Reserve, C-class and seasonal storage occupy the rear and connect to the main replenishment aisle.
Pick frequently ordered items efficiently	A-class and B-class picking sit in forward, accessible areas close to consolidation and packing.
Dispatch completed orders	Packing leads directly to outbound staging and three outbound docks.

Material flow

Normal flow: Receiving and QC -> put-away -> reserve or picking storage -> order picking -> consolidation -> value-added services when required -> packing -> outbound staging -> dispatch.

Exception flow: Returns -> inspection and quarantine -> approved stock release or separate disposition. This keeps returned or rejected garments outside the normal available-stock flow.

3 Exact dimensions and area allocation

The enclosed building footprint is 60 m x 50 m, giving 3,000.00 m². The functional zones total 2,370.07 m², or 79.00% of the footprint. Walls, circulation and clearances occupy the remaining 629.93 m², or 21.00%. External truck yard, loading apron, parking and landscape areas are outside the 3,000 m² enclosed footprint.

Zone	Dimensions m	Area m ²
C-class slow-moving	16.0 x 14.4	230.40
Reserve pallet storage	29.0 x 14.4	417.60
Seasonal flexible	13.0 x 14.4	187.20
Garment-on-hanger	13.4 x 14.0	187.60
Returns quarantine	13.4 x 7.0	93.80
B-class picking	12.0 x 12.0	144.00
A-class fast pick	19.5 x 12.0	234.00
Order picking staging	12.0 x 8.5	102.00
Consolidation	10.0 x 8.5	85.00
Value-added services	9.1 x 8.5	77.35
Packing area	19.5 x 5.0	97.50
Receiving and QC	20.4 x 9.4	191.76
Outbound staging	19.5 x 4.4	85.80
MHE charging maintenance	7.4 x 8.0	59.20
IT server	7.4 x 4.0	29.60
Office administration	7.4 x 9.5	70.30
Staff facilities	7.4 x 7.5	55.50
Staff access	7.4 x 2.9	21.46

Functional-zone total: 2,370.07 m². Circulation, walls and clearances: 629.93 m². Enclosed total: 3,000.00 m².

4 How to calculate future operational requirements

The design already has fixed geometry. The following formulas show how to test whether the geometry can handle a future operating scenario. Insert real operational data where brackets appear.

Reserve pallet storage

Required pallet positions = peak reserve inventory in pallets / target storage utilisation. Reserve storage area = required pallet positions x area per pallet position including rack access and aisles.

For example, the 417.60 m² reserve zone should be tested using the selected rack configuration, pallet dimensions, number of levels, rack aisle requirement and target utilisation. Do not call it a verified number of pallet positions until these inputs are confirmed.

Receiving and outbound staging

Receiving staging positions = peak inbound pallets per hour x average receiving dwell time in hours. Outbound staging positions = peak dispatch-wave pallets x average staging time. Add the handling and circulation space needed around those positions.

Forward picking

Pick-face requirement = number of SKUs in a velocity class x required facings per SKU. A-class products receive the most accessible forward-pick locations because they create the highest pick frequency. B-class products receive intermediate locations. C-class products can sit farther from dispatch because they are accessed less often.

GOH, packing and processing

Required rail length = peak hanging-garment quantity / garments per metre of rail. Packing area = number of packing stations x area per station plus queue and circulation allowance. Consolidation area = concurrent orders waiting to be combined x average order footprint.

Dock requirement

Number of dock positions = peak hourly vehicle or pallet workload x average handling time / available dock time. The five-dock arrangement is a design choice in this concept and should be checked against real arrival and departure schedules.

5 Zone-by-zone justification

Receiving and QC 191.76 m²

Located at the front-left beside the two inbound docks. It accommodates temporary inbound staging, inspection activity and a QC workstation before put-away. Test it against peak inbound pallets per hour and inspection time.

Returns and quarantine 93.80 m²

Placed immediately behind receiving. This allows returns or damaged garments to be inspected and held separately before approval for stock release.

Reserve pallet storage 417.60 m²

The largest zone because it holds bulk carton inventory and feeds the forward-picking areas. Rear-centre placement links it to the replenishment aisle.

C-class slow-moving 230.40 m²

Rear-left location uses accessible but less prime space for products with lower picking frequency.

Seasonal flexible 187.20 m²

A dedicated buffer for collection launches, promotions and temporary stock changes. It equals 6.24% of gross warehouse area.

GOH 187.60 m²

Separate hanging storage protects garments that should not be folded or compressed. Final rail quantity depends on garment profile and rail density.

A-class 234.00 m² and B-class 144.00 m²

A-class receives the largest forward-picking space because fast-moving SKUs need more pick faces and frequent replenishment. B-class needs accessible, but less premium, space.

Order staging 102.00 m², consolidation 85.00 m² and VAS 77.35 m²

These adjacent zones support orders in progress, the merging of items from multiple pick areas, and tagging or repacking when required.

Packing 97.50 m² and outbound staging 85.80 m²

Packing is directly above outbound staging, reducing movement after orders are packed and before they are loaded through the outbound docks.

MHE and support rooms

Charging/maintenance, IT, office, staff facilities and staff access occupy the right-side strip so support activity stays out of the core material flow.

6 MHE pedestrian and access check

The layout should distinguish between spaces used by people and spaces used by material-handling equipment. Small gaps between process zones are not automatically forklift aisles. Forklifts should use the designated replenishment and MHE routes shown in the model.

Forklift check to complete

Input	What to record	Why it matters
Forklift envelope	Overall width, length, mast height and turning radius from the manufacturer data sheet.	Determines travel and turning envelope.
Load envelope	Pallet dimensions, load overhang and carried-load height.	A loaded truck requires more space than the truck alone.
Measured route	Minimum clear width and turning locations along every designated MHE route.	Confirms that the actual layout suits the selected truck and load.
Operating rule	One-way or two-way travel, speed, reversing controls and times of pedestrian access.	Changes the effective clearance requirement.

For this project, use the selected 4 ft wide and 10 ft long forklift as the planning equipment. Show a measured FreeCAD screenshot of the actual designated routes and turning areas. State that the route review confirms available clearance for the chosen operating method only after the actual manufacturer turning specification is applied.

Pedestrian and dock safety logic

The central staircase and the right-side staff entrance provide pedestrian access without requiring staff to use loading doors. The plan marks a pedestrian route beside the support strip and MHE movement routes in the operating area. A final design should add physical barriers or controlled crossings at conflict points and check the dock edge, dock plate and truck-restraint arrangements.

The design principle follows OSHA guidance requiring safe mechanical-handling clearance at aisles, docks, doorways and turns, and HSE guidance that pedestrians and vehicles should be separated wherever possible. These are planning benchmarks; applicable Sri Lankan rules and the final risk assessment govern the actual installation.

8 Evidence checklist and useful KPIs

Collecting the following information will allow you to convert the concept into a more defensible facility design study.

Evidence to collect	How it strengthens the design
Forklift manufacturer sheet	Validates turning radius, loaded aisle needs and mast height.
Pallet/carton/GOH dimensions	Converts zone area into realistic storage capacity.
ABC analysis and active SKU count	Justifies the relative A, B and C pick-zone sizes.
Peak inbound/outbound volumes	Tests receiving, staging and dock sizes.
Return rate and quarantine dwell time	Tests the returns/quarantine allocation.
Packing labour and workstations	Tests packing, VAS and consolidation space.
FreeCAD measurement screenshots	Demonstrates MHE routes, stair access, dock arrangement and zone dimensions.

KPIs to measure after implementation

KPI	Definition
Storage utilisation	Occupied usable storage capacity / available usable storage capacity x 100.
Throughput	Units, order lines or pallets processed per day or hour.
Order cycle time	Time from order release or receipt to dispatch.
Picking productivity	Picks or order lines completed per labour-hour.
Travel distance	Metres travelled per order, pick or pallet movement.
Dock utilisation	Active dock time / available dock time x 100.
Order accuracy	Correct orders / total orders x 100.

Sources used for design principles

University of Moratuwa, LT3050 Planning and Design of Transport and Logistics Facilities, Warehouse Design lecture material supplied for the case study.

Occupational Safety and Health Administration, 29 CFR 1910.176 Handling materials - general.
<https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.176>

Health and Safety Executive, Separating pedestrians and vehicles.
<https://www.hse.gov.uk/workplacetransport/separating.htm>

Health and Safety Executive, Pedestrians. <https://www.hse.gov.uk/workplacetransport/factsheets/pedestrians.htm>